

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a6bc611a30c588b90.jsonl`  
- SHA-256: `42fef61c3e5d7ef503f4d5c3c2f1ecdeca2ad9a99a730109fedea08e0875f97c`  
- Source modified: `2026-06-16T03:21:08+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:24.335Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://ar5iv.labs.arxiv.org/html/2204.01899>

****Title:**** MonoTrack: Shuttle Trajectory Reconstruction from Monocular Badminton Video (CVSports/CVPRW 2022)

****Found via:**** Shuttle/ball trajectory ? rally/point segmentation in racket sports search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:26.089Z)

I'll fetch the source page and extract relevant claims.

3. user (2026-06-16T03:20:42.315Z)

MonoTrack: Shuttle Trajectory Reconstruction - Key Claims

(1) Trajectory Reconstruction Method & Accuracy

****Reconstruction Approach:****

"Our system integrates badminton domain knowledge such as court dimension, shot placement, physical laws of motion, along with vision-based features such as player poses and shuttle tracking."

The system uses physics-based modeling: "The 3D trajectory, $\mathbf{p}(t)$, can simply be approximated by a particle under drag" with nonlinear optimization of reprojection error against 2D tracked coordinates.

****Reported Accuracy:****

- ****Synthetic data:**** "On average, adding the priors improves the error from 14.9 cm to 8.0 cm"
- ****Real data (2D reprojection):**** Using ground truth labels, errors ranged 8.1-9.8 pixels; end-to-end system achieved 23.3-37.1 pixels (approximately 5% of image resolution)
- ****Shuttle detection:**** "88.6% accuracy" (improved from prior 84.0%)

(2) Rally/Point Segmentation & Stroke Detection

****Hit Detection Architecture:****

"A recurrent model that leverages the previously computed court layout, poses, and shuttlecock positions...to predict hits" using "two GRU layers that takes 12 frames at a time."

****Method:**** "Identifying the shots of a rally is critical...A shot happens when a player hits the shuttle with her racket, and ends right before the opposing player hits the shuttle."

Three-way classification: "no hit," "near player hit," "far player hit" per frame.

(3) Reported Accuracy Numbers

****Hit Detection Performance (Table 1):****

- HitNet with optimization: ****94.3% recall, 89.7% accuracy, 94.9% precision, 0.946 F1****
- Ablation study shows court, pose, and shuttle all contribute significantly

****Overall Shot Accuracy:****

"The network...can reach 86.3% accuracy" without optimization; optimization substantially improves results.

(4) License & Code Availability

No explicit license statement or code/weights availability mentioned in the paper.

(5) Publication Venue & Date

****ArXiv preprint:**** arXiv:2204.01899, published April 2022

****Authors:**** Paul Liu (Stanford University), Jui-Hsien Wang (Adobe Research)

4. assistant (2026-06-16T03:20:49.638Z)

I have the extracted content. Let me assess and structure the findings.

Source quality: This is a primary research paper (arXiv preprint, CVSports/CVPRW 2022) authored by Paul Liu (Stanford) and Jui-Hsien Wang (Adobe Research). Primary source.

The paper is directly relevant to Dimension 2 (shuttle-trajectory ? rally signal processing) ? specifically the hit/stroke detection that converts trajectory into per-shot segments, which is the building block for distinguishing in-play strokes from dead-time.

5. user (2026-06-16T03:21:02.823Z)

Structured output provided successfully

6. assistant (2026-06-16T03:21:08.565Z)

Extracted and verified claims from MonoTrack (arXiv:2204.01899, CVSports/CVPRW 2022, Liu & Wang).

Key takeaway for the research question: this is a primary source directly relevant to Dimension 2. Its central contribution applicable to the over-merge problem is HitNet ? a GRU-based recurrent model over 12-frame windows that does per-frame 3-way hit classification (no hit / near-player hit / far-player hit) fusing court layout + pose + shuttle position, reporting 0.946 F1 / 89.7% accuracy. Hit/contact events are the natural rally-internal boundary signal that could split the merged mega-windows our pipeline produces. The physics-of-motion (drag-constrained) trajectory priors (14.9cm?8.0cm error reduction) support fps-invariant trajectory features. Caveat flagged as unverified: the paper text states no explicit license, so commercial-clean reuse under the MIT/Apache/BSD constraint requires separate repo-license verification.